

```

122.         #      |    0    | DICE_HEIGHT (5)
123.         #      |      0    | |
124.         #      +-----+ v
125.         #      <----->
126.         #      DICE_WIDTH (9)
127.         topLeftX = left
128.         topLeftY = top
129.         topRightX = left + DICE_WIDTH
130.         topRightY = top
131.         bottomLeftX = left
132.         bottomLeftY = top + DICE_HEIGHT
133.         bottomRightX = left + DICE_WIDTH
134.         bottomRightY = top + DICE_HEIGHT
135.
136.         # Check if this die overlaps with previous dice.
137.         overlaps = False
138.         for prevDieLeft, prevDieTop in topLeftDiceCorners:
139.             prevDieRight = prevDieLeft + DICE_WIDTH
140.             prevDieBottom = prevDieTop + DICE_HEIGHT
141.             # Check each corner of this die to see if it is inside
142.             # of the area the previous die:
143.             for cornerX, cornerY in ((topLeftX, topLeftY),
144.                                     (topRightX, topRightY),
145.                                     (bottomLeftX, bottomLeftY),
146.                                     (bottomRightX, bottomRightY)):
147.                 if (prevDieLeft <= cornerX < prevDieRight
148.                     and prevDieTop <= cornerY < prevDieBottom):
149.                     overlaps = True
150.         if not overlaps:
151.             # It doesn't overlap, so we can put it here:
152.             topLeftDiceCorners.append((left, top))
153.             break
154.
155.     # Draw the dice on the canvas:
156.
157.     # Keys are (x, y) tuples of ints, values the character at that
158.     # position on the canvas:
159.     canvas = {}
160.     # Loop over each die:
161.     for i, (dieLeft, dieTop) in enumerate(topLeftDiceCorners):
162.         # Loop over each character in the die's face:
163.         dieFace = diceFaces[i]
164.         for dx in range(DICE_WIDTH):
165.             for dy in range(DICE_HEIGHT):
166.                 # Copy this character to the correct place on the canvas:
167.                 canvasX = dieLeft + dx
168.                 canvasY = dieTop + dy
169.                 # Note that in dieFace, a list of strings, the x and y
170.                 # are swapped:
171.                 canvas[(canvasX, canvasY)] = dieFace[dy][dx]
172.
173.     # Display the canvas on the screen:
174.     for cy in range(CANVAS_HEIGHT):
175.         for cx in range(CANVAS_WIDTH):
176.             print(canvas.get((cx, cy), ' '), end='')

```